

Plant Water Management Experiments 5 & 6 on ISS: Hydroponics and Ebb and Flow

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Crop production in microgravity will be important to provide whole food nutrition, dietary variety, and psychological benefits to astronauts exploring deep space. Unfortunately, terrestrial plant watering methods face significant challenges when applied aboard spacecraft due to rogue bubbles, ingested gases, ejected droplets, and various unstable liquid interface configurations that arise in microgravity environments. To address these issues, we present the recent results of the Plant Water Management (PWM 5 & 6) technology demonstrations conducted on ISS, where recirculating hydroponic and ebb and flow watering processes are studied using engineered root modules varying solution flowrates, serial and parallel channel fill levels, and analog root densities. The microgravity demonstrations employ plumbing elements that exploit the near absence of body forces to achieve passive bubbly flow aeration, bubble separation, gas diversion, bubble mergers and migration, droplet capture, nutrient/water delivery, and the use of plant root geometry for complete microgravity fluid phase separations at different stages of growth. Practical demonstrations of system priming, start-up, shut-down, draining, and plant installation and changeout are highlighted. Such advances offer a variety of plug-and-play solutions for effective plant watering in low- and variable-gravity environments, despite the challenging wetting properties of contaminated aqueous nutrient solutions.

Acronyms and Nomenclature

AIS	=	Air Injection Syringe	AOS/LOS	=	Acquisition/Loss of Signal
BD	=	Bubble Diverter	BS	=	Bubble Separator
CV	=	control valve	E&F	=	Ebb and Flow Technique
MWA	=	Maintenance Work Area	NFT	=	Nutrient Film Technique
PTFE	=	Teflon	PV	=	pinch valve
PWD	=	Potable Water Dispenser	R1-R4	=	plant root model types
Res	=	capillary reservoir	SC	=	stopcock
SH	=	superhydrophobic	SHic	=	superhydrophilic
TC	=	Test Channel	V	=	Vol.: TC, Res, Syringe, Bubble
WT	=	Water Trap	Q	=	pump flowrate

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I. Introduction

PLANT growth habitats aboard spacecraft, moon, and Mars must address the challenges of routine, robust, and automated water delivery despite the significant reduction in gravitation. Of many watering methods available, NFT Hydroponics and Ebb and Flow methods are pursued herein with a special focus on the microgravity environment. From a spacecraft efficiency point of view, such methods are clean, cleanable, productive, without substrate, automated, provide semi-passive control, and more. However, numerous persistent challenges abound due to the numerous, normally mundane, capillary fluidic phenomena peculiar to the microgravity environment. These include unearthy capillary fluidics over large length scales: (1) unsteady developing transitional multiphase flows, (2) the instability of bubbles, films, rivulets, jets, geysers, and drops, (3) poor, variable, and hysteretic wetting, (4) contact line pinning, (5) complex metastable free surface geometries, (6) stability of highly parallelized inertial-capillary channels, and (7) practical proof of the durability of ‘permanent’ solutions. The presence of gravity in terrestrial plant watering seems to take care of everything without even a second thought from the gardener. Our concern here is the microgravity fluid physics of plant watering aboard spacecraft, which often reduces to the management of low-g bubbles and drops.

In this paper, following a brief overview of PWM Hydroponics, we present a summary of the PWM-5 & 6 technology demonstrations on ISS during 2024. We provide brief descriptions of flight hardware, crew procedures, data reduction, and status of the PWM data archive. Preliminary assessments of select capillary fluidic device performance are highlighted.

II. PWM-Hydroponics Overview

The Plant Water Management Hydroponics experiment (PWM-Hydroponics, or PWM) is pursued by NASA as a series of low-cost fast-to-flight technology development investigations to demonstrate the feasibility of the general approach and to advance the readiness level of the microgravity hydroponics technology. Background for the PWM experiments is provided in part in Refs. 1 and 2. In Refs. 3 and 4, the ISS PWM-Hydroponics 3 & 4 experiments are reported for 6 8-hour crew days during the March-July 2021 time frame. The central objectives of those demonstrations concerned system priming, start-up, stable single and two-channel parallel flow, impact of single and multiple simple wicking and evapotranspiring plant models varying size and complexity, response to varying fill levels, flowrates, ease of plant insertion and removal, shut down, and more. Specific tests to identify the limits of stable operation as well as the passive mitigation, diversion, and separation of aerating bubbles were reported with over 400 individual tests performed and summarized.

The successful demonstrations of PWM 3 & 4 lead to the advanced Test Channel and Gas Diverter designs of PWM 5 & 6, adding hydroponic and ebb and flow demonstrations with higher fidelity plant root models, cavalier priming, controlled aeration, novel gas separation, water trap and separation, zoomed high-resolution 4K video with 2 FOVs, and more. As with PWM 3 & 4, a raw and reduced video data archive with hyperlinked clips of and links to the individual test set-points and video events for PWM 5 & 6 will be made publicly available on the NASA PSI database (<https://psi.nasa.gov>).

The PWM-Hydroponics hardware is designed for safety and simplicity of use. At the beginning of each operation, the hardware is manually unstowed and assembled on the portable Maintenance Work Area (MWA, i.e., workbench) aboard the ISS. All tests are performed by crew in the open cabin of the ISS, after which the assembly can be quickly drained, disassembled and stowed, with all quantitative measures recorded via HD and/or 4K video cameras (e.g., flowrates, fill levels, interface configurations, bubble distributions, velocities, etc.).

III. PWM 5 & 6

NASA authority to proceed on the PWM 5 & 6 technology demonstration began with project kick-off meeting June 29, 2022. Despite the ‘Tech Demo’ designation, the project addresses numerous applied microgravity fluid physical phenomena rich in practical cross-cutting exploration value beyond plant watering (i.e., fuels, cryogenics, coolants, ECLSS fluids, water processing, bio-fluidics, medical fluids, hygiene, habitats, etc.). Thus, the hardware was designed with quantitative measures of rare critical low-g capillary fluidic phenomena in mind. This was to be accomplished employing transparent plumbing elements and digitized HD video data. The hardware was designed, fabricated, flight certified, and delivered October 4, 2023, for launch to ISS November 10, 2023 on SpX-29. Greatly simplifying and speeding the flight qualification process, the decision to employ NutraSweet fruit punch as the test fluid was made early. The thermal-fluid and wetting properties are nearly identical to the nutrient solutions used in true hydroponics, but fruit punch is TOX-0 allowing the experiments to be conducted on the MWA in the open cabin

of the ISS. A 48-hour limitation is placed on the wetted experiments due to a potential bio-growth hazard. After 48 hours, the experiments are terminated, and the hardware is ‘wet-trashed.’

A. Hardware Overview

Labelled images of the PWM 5 & 6 hardware are provided in Figure 1. The hardware consists of a hydroponic Test Channel (TC), a tubing harness, and a peristaltic pump to circulate liquid through the system. The capillary fluidic elements of the TC, Reservoirs (Res), Bubbles Separator (BS), Water Trap (WT), and Bubble Diverter (BD) are 3D printed Formlabs Clear® parts and the 3/16" ID Tygon® tubing is connected using Luer Lock fittings. For PWM-5, two 120 mL syringes are used to prime the system, adjust liquid levels during operations, and make up for liquid losses to evaporation. A 30 mL gas syringe is filled with cabin air and used to inject bubbles into the system during specific tests. The liquid for the tests is an in-flight re-constituted NutraSweet®-based tropical Fruit Punch prepared by the crew prior to the tests. The thermophysical properties of the beverage are similar to typical plant nutrient solutions with surface tension $\sigma \approx 0.064$ N/m, density $\rho \approx 1010$ kg/m³, dynamic viscosity $\mu = 0.0012$ kg/m·s, and channel polymer contact angle $\theta \approx 40 \pm 22^\circ$. The variable-speed peristaltic pump provides flow in the range of 0.8-5.1 mL/s (for 3/16" ID pump head tubing). The pump flow delivery rate is nominally manually adjusted by the crew using the dial on the pump, with precise values determined post-flight by image processing the video footage via FFT analysis of the pump head rotation rate and applying 0.857 mL/cycle for the 3/16" ID pump head tubing employed. Pre-flight and in-flight Pump calibrations are used to spot check flowrate accuracy. All components are mounted to 3D printed ULTEM™ back and base plates, which are then secured to the MWA with hook-and-loop fasteners.

The hardware may be configured to establish a variety of flow configurations through the plumbing elements and TC during operations. The TC is positioned for distortion-free observation of the flow behavior. In all cases, the TC flow direction is left to right. Two identical PWM-5 units (5 and 5b) and one PWM-6 unit were shipped to ISS. The devices include at least 10 capillary fluidic devices, and at least 14 capillary fluidic fittings. The fluidic devices include TCs, Aerator, Reservoirs, Peristaltic Pump, Air Injection Syringe (AIS), BS, Control Valve (CV), WT, and BD and are designed with passive capillary control in mind. The fluidic fittings are terrestrial COTS items that include the supply drink bag, syringes, fittings (unions, elbows, tees, wyes, expansions, contractions), pinch valves (PV), and CVs that do not consider capillarity in their function. In the presence of capillary two-phase flows, all such plumbing elements play a role in the developing flow regimes. As inadvertent examples, the syringes can serve as bubble generators, simple elbow fittings can shatter bubbly flow regimes, unions can hold-up bubble trains prematurely converting bubbly flows to gas slug flows, and the peristaltic pump can convert regular bubble trains into irregular bubbly flows, or even a ‘froth’ at high flowrates.

Solid models of four of the primary capillary devices of PWM 5 & 6 are provided in Figure 2: TC, BD, BS, and WT. Characteristic dimensions are included. The differences between PWM-5 and PWM-6 are that PWM-5 employs one TC and two Reservoirs, while PWM-6 employs two TCs and no Reservoirs. PWM-6 pursues the stability of parallel inertial-capillary hydroponic channels as a function of root model. Both PWMs are capable of dual Hydroponic and E&F demonstrations with targeted performance testing on TC, Aerator, BS, WT, and BD. Brief descriptions of these devices are provided herein.

Test Channel (TC). Test Channel perspectives with dimensions are provided in Figure 2a. The 7 cm high, 16 cm long TC is a 20° ($\alpha = 10^\circ$) included angle open wedge channel with centered tapered cusp drain at its base, continuous 90° pinning edge, and superhydrophobic (SH) living hinge lid with 3 ports for plants. The TC holds 113 mL when filled flush to the pinning edge. The lower inlet for the TC is aligned with the cusp and the upper inlet is 23.6 mm higher. The non-wetting inner lid surface is a SH-PTFE monolithic screen (not a coating) to allow for translucent visualization. The lid is secured with clear tape serving as a living hinge for ease of plant placement and removal. The TC is SLA printed of flight certified Formlabs Clear® Resin. Upper and lower inlets are used, but only a single lower outlet. As previously demonstrated on ISS^{1,2}, and building upon enhancements from the Space Cup⁵, the wedge geometry provides excellent stability for flow along the wedge with the passive capability to drive bubbles toward the free surface where they merge and leave the flow. Thus, the TC provides a measure of passive bubble separation for the system. This mechanism is promoted further by the presence of plant roots which trap bubbles, enhancing mergers, and driving them out of the channel with increased efficiency. The TC dimensions balance the desire for large fluid volume and elevated flowrates (up to 10 mL/s) with adequate stability to crew induced perturbations ($\sim 10^{-2}g_0$) resisted by hysteretic and/or pinned contact lines due to the partial wetting conditions of the polymer-fruit punch system with contact angles $30^\circ < \theta < 80^\circ$. The presence of plant roots enhances liquid stability by reducing characteristic free

surface dimensions and increasing viscous damping. For dense root packings, the TC can serve as a perfect passive bubble separator.

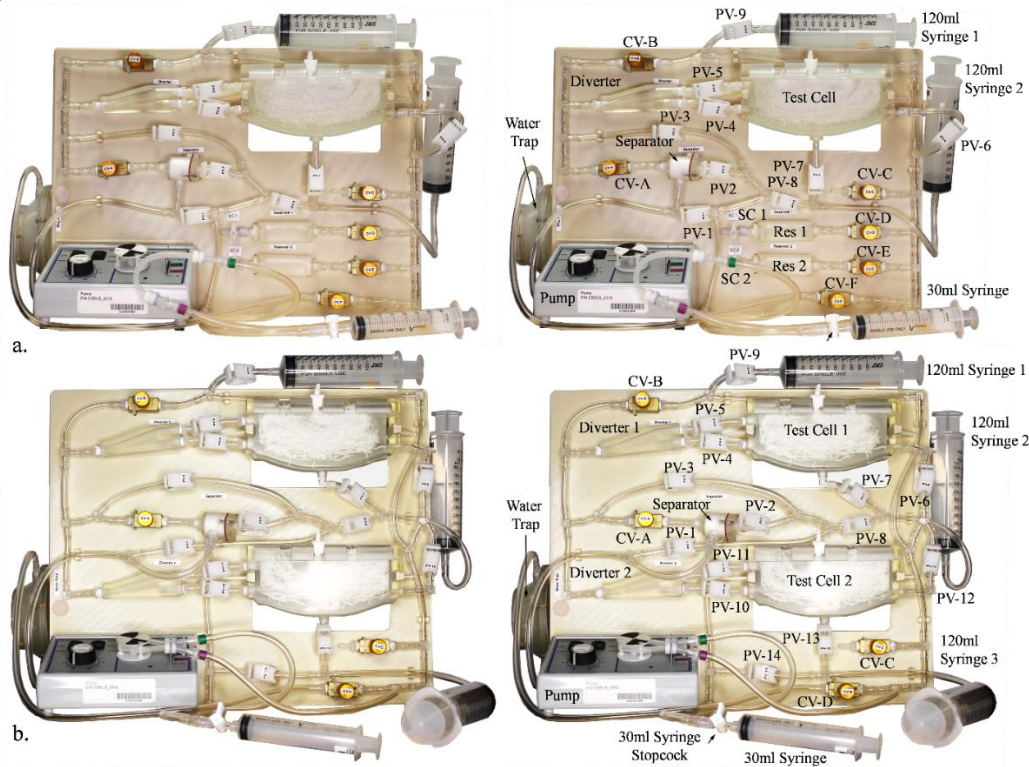


Figure 1. Images of dry flight hardware with and without labels for a. PWM-5 (equivalent to PWM-5b) and b. PWM-6 with Root R4 installed in Test Channel (TC). Here ‘Separator’ is BS, ‘Diverter’ is BD, and ‘30ml Syringe’ is AIS.

Bubble Diverter (BD). The BD is best described as a gas diverter⁶ (Fig. 2b). The device splits the incoming flow into single phase liquid flow into the lower TC inlet, while a higher gas concentration two-phase flow is directed into the upper TC inlet. The TC inlets establish similar downstream pressures, and the bubble separation mechanism is managed by a bubble point-limited sorting mechanism of the integrated diverter screen and inlet bubble diameter distribution. For bubble diameters $< \delta$ and device pressure drop $\Delta P_{BD} < P_{\delta BP}$, 100% liquid enters the lower TC inlet port. The increased density of bubbles directed into the TC inlet port merge quickly and are driven immediately to the free surface where the far majority escape the channel flow reducing the need for the channel to separate the bubbles, and reducing the potentially negative impacts of bubbles deep within the plant roots which tend to lift the roots out of channel³. The BD is capable of diverting a variety of two-phase flow regimes such as sparse bubble, bubbly, bubble slug, and gas slug flows). The BD is currently TRL 8 with devices currently employed in cross-cutting applications aboard ISS (i.e., urine pretreatment concentration measurement device where only liquid is to be sampled from a sparse bubble two-phase flow).

Bubble Separator (BS). The BS is more specifically described as a Superhydrophobic Multiplexed Inertial Filter, Figure 3c. The Multiplexed Inertial Filter was first patented for low-g droplet/mist capture applications aboard spacecraft⁷ but is reformatted for PWM using SH materials to achieve 100% bubble separations from aqueous solutions⁸. Not shown in Figure 2c, the BS employs numerous parallel SH helical passages to centrifuge and adhere bubbles to the porous packing allowing the locally pressurized gas to escape the otherwise uninterrupted liquid flow. Complete phase separations are possible with such an easily deployed in-line device and the potential benefits to de-bubbling liquids in space are compelling.

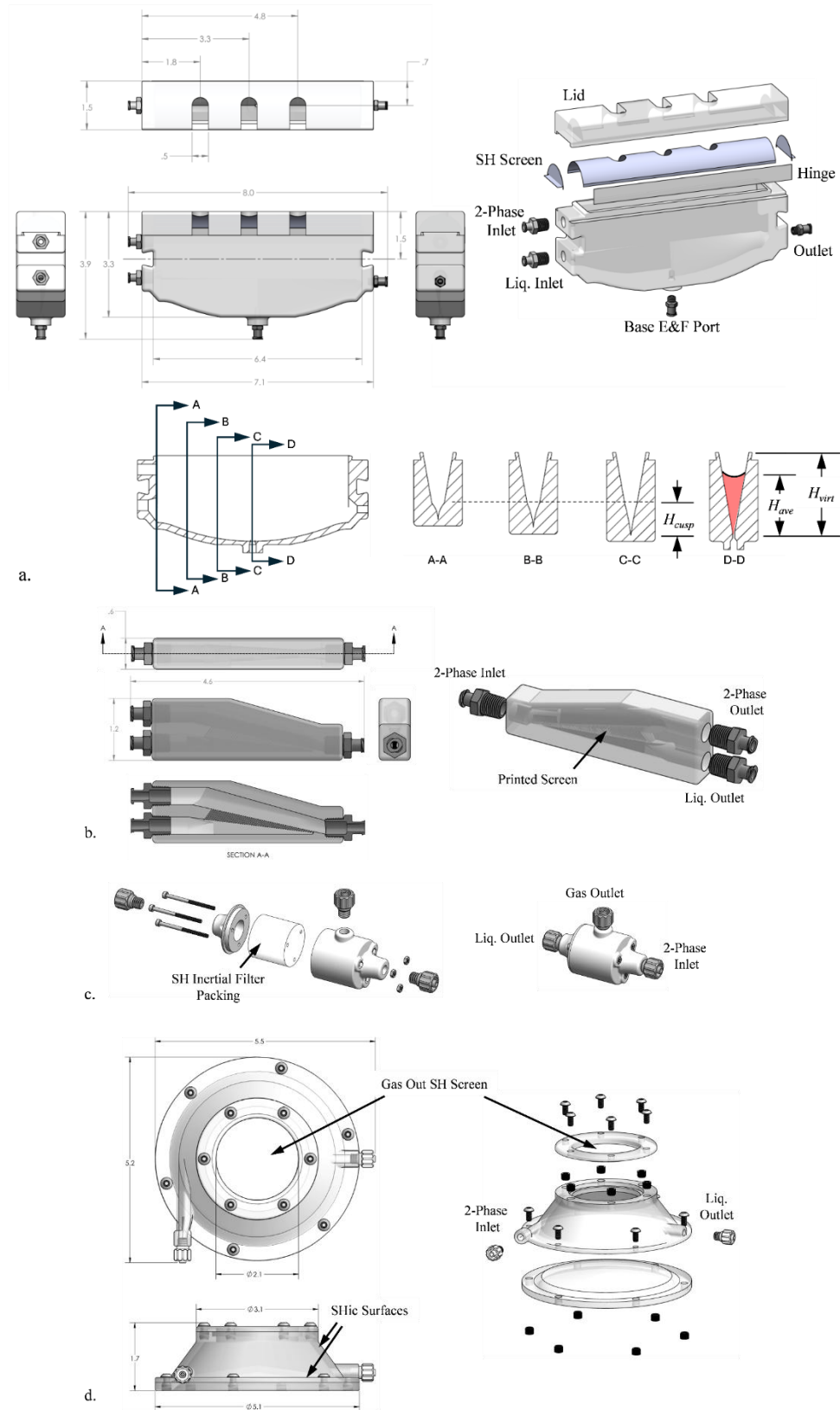


Figure 2. Solid model details of four primary capillary devices of PWM 5 & 6: a. TC, b. BD, c. BS, and d. WT. (Dimensions in inches)

Water Trap (WT). More appropriately described as a low-g Capillary Coalescing Filter in Figure 2d, the WT is employed with PWM as a redundant device to assure that no liquid escapes from the BS gas outlet line into the ISS cabin. During BS limit tests it is expected that the BS is pushed beyond the limit of liquid escape through the gas outlet line. Such widely varying liquid content two-phase flow is captured and held by the superhydrophilic (SHic) circular interior corner of the WT (conical sidewalls and base), while allowing bubbles to escape into the core and out of the WT through a central circular SH-PTFE screen. The interior corner of the device performs a similar bubble separating function as the open wedge TC, but with improved performance due to the perfectly wetting SHic Rayon-lined walls of the WT wedge and slight centrifugal forces generated by the circumferential flow. The WT approach is patent pending⁹ and has found cross-cutting applications (i.e., primary separation function of a spacecraft wet vac). The WT functions as a passive phase separator or a water trap depending on whether the liquid is or is not drained during operation.

Aerator. The aerator consists of a simple Tee fitting through CV-C just downstream of the TC on the low-pressure side of the loop, Figure 1. More elaborate capillary venturi aerators were originally designed and built for this function, but the simple COTS Tee junction performs adequately, and with such a small form factor. The sensitive control valve CV-C (1/8" Deltrol Needle Valve, EN10B) allows the crew to regulate the air flowrate into the loop. The added degree of freedom of flowrate adjustments at the pump establishes a wide variety of aerating two-phase flows. The capability of generating aerating bubbles and then safely removing them essentially eliminates concerns of historically hypoxic water supplies aboard spacecraft. Because the pump volumetric flowrate is known and the gas flowrate Q_g may be measured from the video data, the liquid flowrate may be found from $Q_l = Q_{pump} - Q_g$, from which the flowrate ratio may be found, $\phi = Q_g/Q_l$. For occluding flows, for fixed Q_{pump} , increasing the aeration rate does not change the liquid velocity, but reduces the liquid flowrate, increasing the flowrate ratio. We note that increasing aeration decreases liquid superficial velocity ($= Q_l/A_{tube}$).

Reservoirs (PWM-5 and -5b). Dual ≈ 20 mL Reservoirs are connected to the suction side of the loop just upstream of the Pump for the PWM-5 and identical PWM-5b hardware, Figure 1a. The two reservoirs emulate plant nutrient solution and water supply. By opening the Reservoir breather stopcocks (SC), adjusting the Reservoir control valves CV-D and CV-E determines the routine steady passive injection of simulated 'nutrient solution' and 'water supply.' Several demonstrations of this straightforward capability varying injection rates and volumes are performed. Automation of the process is a natural direction for the development of the technology.

Syringes. The 120 mL Syringes are filled directly from the onboard ISS fruit punch drink bags. They are then connected to the PWM 5 & 6 conduit and employed to prime the system, adjust TC Hydroponic flow fill levels, conduct E&F tests, and drain the system. It is difficult to withdraw bubble-free liquid from a drink bag and the crew often applies centrifugal forces before bleeding gas out of the 120 mL Syringes and connecting to the loop. However, it is possible to conduct bubble-free syringe primes and drains of the PWM TCs and loop with a concerted effort not to exceed the ingestion limits (velocities) of the in- and out-flow liquid. But such manually controlled flows require continued concentration from the crew and continued patience from the science team on the ground. Because the PWM 5 & 6 hardware contains redundant passive bubble separating devices, recovery from any inadvertent bubble ingestion is automatic. The impact of this 'feature' of the PWM hardware subconsciously acts on the science team to relax concerns about bubbles in the syringes and loop and press ahead with faster more cavalier syringe fills and drains. It was routinely heard in the control room that 'we can determine the liquid content in the bubble-filled 120 mL Syringes by optically measuring the gas contents post-flight.' Unfortunately, this is not always possible.

The 30 mL Syringes (Air Injection Syringe, AIS) attach to the positive pressure side of the loop just downstream of the pump. Using these air-filled syringes, highly regulated bubbly flows and bubble trains may be injected into the loop without being churned by the peristaltic pump. In such cases, the pump flowrate is the liquid flowrate, and the injection rate is the gas flowrate. Thus, $Q_{tot} = Q_{ginj} + Q_{pump}$ and $\phi = Q_{ginj}/Q_{pump}$. For occluding flows, the liquid velocity increases with air injection, but the liquid flowrate and superficial velocity remain constant. A two-way stopcock attached to the 30 mL Syringe allows for various 'clean' bubble injections serving as a second aerator. However, at high pump speeds, fittings downstream of the gas injection location still managed to disrupt the clean bubble flow regime injected.

Modes of Operation. The somewhat entangled plumbing of the PWM Hydroponic hardware enables the variety of loop functions necessary to meet the technology development requirements of the investigation. Six of the numerous modes of the PWM hardware operation are sketched in Figure 3 for PWM 5 & 6. Replacement of all four root models

is assumed as well as Hydroponic tests with no root models. Duplicate circuits are not shown for PWM-6. Modes with Reservoirs engaged are not shown. The loops include (1) E&F operations from 120 mL Syringes 1 and 2 through TC base, lower inlet, and outlet (ref. Fig. 3a-b), (2) Hydroponic flow with and without Aeration, Gas Injection, and Bubble Diversion (ref. Fig. 3c), (3) Hydroponic flow with and without Aeration, Gas Injection, Bubble Separation, Water Trap, and Bubble Diversion (ref. Fig. 3d), (4) Water Trap as a variable volume liquid trap and bubble separator (ref. Fig. 3e), and (5) Parallel Hydroponic flow with and without Aeration, Gas Injection, Bubble Separation, Water Trap, and Bubble Diversion (ref. Fig. 3f). The solid red lines symbolize single-phase liquid flow, and the dashed lines symbolize two-phase flow. The latter consists generally of gas slug, bubble trains, or bubbly two-phase flows. Flow regimes are established and reported herein where 100% gas and liquid separations are observed. Other PWM 5 & 6 operational modes can be achieved but were not pursued.

B. Root Models

Early terrestrial tests performed using a variety of plants (i.e., peppers, arugula, etc.) grown in PWM Hydroponic channels were pursued to observe plant root behavior during maturation. (Natural roots were originally proposed for the flight tests but tended to disintegrate when dried.) Results such as depicted in Figure 4 were used to guide the development of the four polyester root models shown in Figure 5. Root model data is listed in Table 1. The multiply cold water rinsed root models mimic the root diameter, length, wettability, and density of the sparse and packed natural roots observed during the terrestrial growth experiments. By touch, root stiffnesses also appeared similar between the wetted synthetic and natural pepper roots. As shown in Figure 6a, the synthetic roots consist of single strands of 100% polyester Coats & Clark Inc. Red Heart® Scrubby Yarn (coconut, Gauge: 4–Medium). The roots of the root models are constructed as depicted Figure 6b-g, with the double-stranded roots separated into single strands, laid out lengthwise and centered, bundled and tied at their center, folded and drawn through a 25.4 mm length of polyolefin heat shrink, and trimmed. The polyolefin heat shrink is then heated for a tight fit while serving as the plant ‘stem’. An approximately 15 mm by 13 mm by 8 mm wedge of poorly wetting closed cell nylon foam is punched for a press fit around the heat shrink stem, as shown completed in Figure 5 and 6g. The foam wedges of the root models slide easily into the slots of the TC lid, 2a.

Table 1. Synthetic root model data: (Fig. 5) Single primary strand diameter is 100 μm , single secondary dead end strand diameter is 458 μm , and each single stand consists of dozens of $\sim 100 \mu\text{m}$ diameter fibers. The single strand root volume and mass per length are $0.382 \pm < 8 \text{ \% mL/m}$ and $0.558 \text{ g/m} \pm < 2$, as determined by submersion and dry mass tests, respectively. The fiber bundles are perfectly wetted by the fruit punch. Note that individual polyester fibers $\approx 100 \mu\text{m}$ and the density for the Polyester was measured at 1.56 g/mL .

Root (R#)	Op.	Nom. Color	#Roots (#/TC)	#Roots/Root Length (#/inch)	#Roots/Root Length (#/inch)	Tot. Root Length (m)	Root Vol. (mL)
R1	sparse	green	3	2/2"	-	0.10	0.1
R2	small	blue	3	4/5"	-	0.25	0.3
R3	medium	black	3	8/5"	-	1.02	1.2
R3	dense	white	1	10/6"	20/24"	13.72	5.2

*Note that ‘R0’ is employed to denote comparative TC flows with no root models installed

IV. Summary of Flight Operations on ISS

Flight operations for the PWM-5 & 6 Hydroponics demonstrations were conducted by eight crew members in three test series: January 16-17, July 16-17, and October 10-11, 2021. Due to the possibility of microbial growth, current ISS safety protocols require the trashing of all wetted components after 48 hours exposure. This requirement guided the decisions on the order of tests performed. ISS flight operations are only summarized herein. Each of the three operations included a hardware setup, system prime, experiment operations, teardown, and stow activity. During system prime and experiment operations, the investigator team had direct audio communication with the crew and real-time downlink video from one or two Sony Camcorders with 1280x720 pixel HD resolution at either 30 or 60 fps and/or 3840x2160 pixel 4K resolution at 60 fps. The crew also took occasional still images throughout the demonstrations using a Nikon D5 DSLR camera. The crew downlinked the locally recorded video and still images to the ground after each operation. The chronological order of ISS flight operations for PWM-5, PWM-6, and PWM-5b are listed in Table 2.

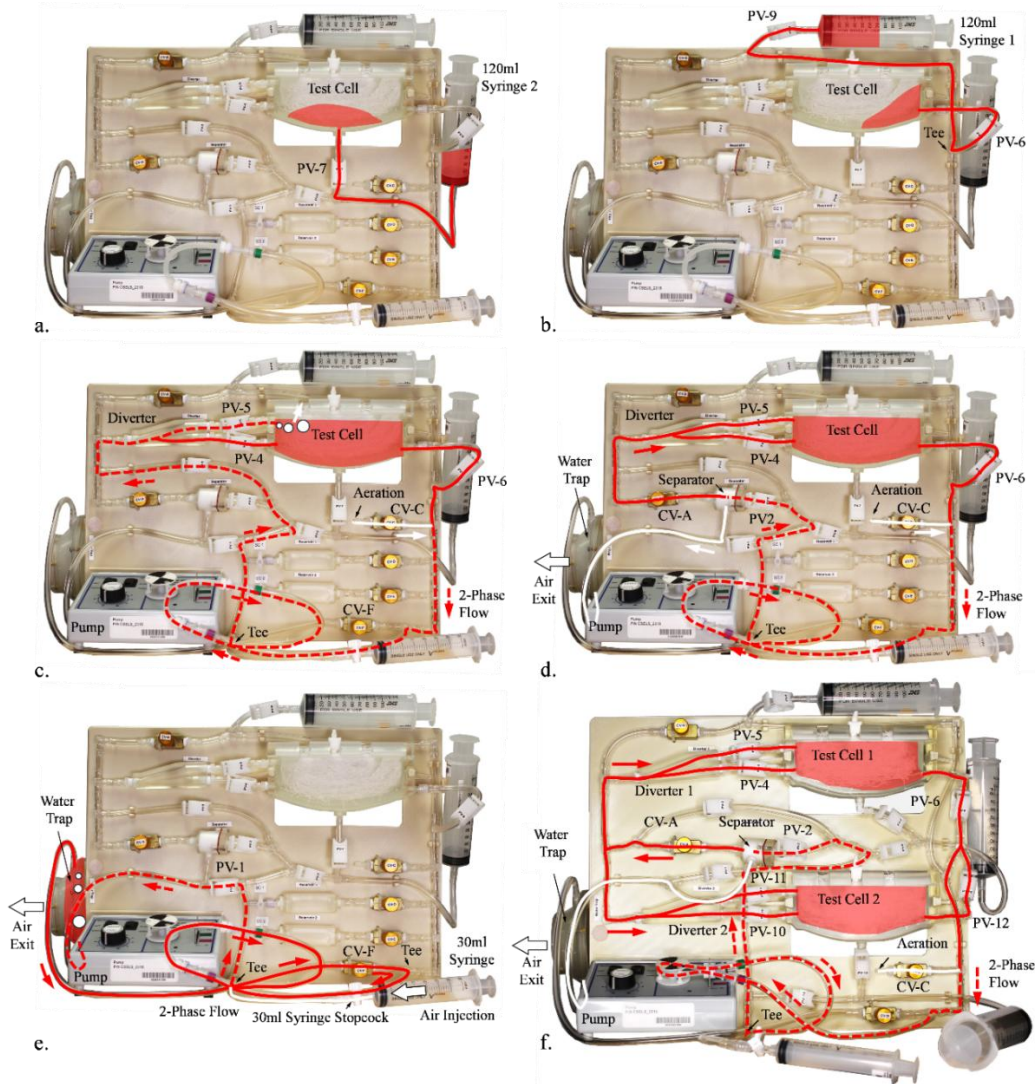


Figure 3. Selection of PWM 5 & 6 modes of operation: PWM-5 E&F operations from 120 mL Syringes through a. TC base and b. TC lower outlet, PWM-5 Hydroponic flow with c. Aeration and Bubble Diversion and d. with Aeration, Bubble Separation, Water Trap, and Bubble Diversion, e. PWM-5 Water Trap with Air Injection and Separation, and PWM-6 Parallel Hydroponic flow with Aeration, Bubble Separation, Water Trap, and Bubble Diversion.

Table 2. Nominal PWM 5 & 6 Flight Operations GMT Chronology completed during 2024 calendar year. Note that hardware set-up was normally proceeded by a 10-20 min space-to-ground crew conference call. All science team space-to-ground communication was performed by co-author L. Torres.

Tech. Demo	Crew	Day 0/1 GMT/hr	Day 2 GMT/hr	Day 3 GMT/hr	Primary Objectives
PWM-5	Moghbeli Furukawa Mogensen	15/01:30	16/08:30	17/08:30	R1-R4: E&F, Single Channel TC Hydroponic flow with and without Aeration, BS, WT, BD, and AIS Long duration runs to increase TRL
PWM-6	Williams Wilmore	177/1:30	178/8:30	179/8:45	R1-R4: E&F, Parallel Channel TC Hydroponic flow with and without Aeration, BS, WT, and BD
PWM-5b	Barratt Benedikt Epps	283/10:30	284/8:20	-	R0-R4: E&F, Single Channel TC Hydroponic flow with focus on high resolution Aeration, BS, WT, and BD with access to 2 science cameras

PWM-5 Performance Summary

The PWM-5 operations sought to demonstrate design advances in all aspects of the PWM-3 and PWM-4 hardware performance including improved TC, TC pinning, TC SH lid, larger volume TC, new E&F capability, new improved BD, new BS, new WT, improved root model variability and fidelity, 4K camera resolution, and more. Numerous ‘long duration’ runs establishing elevated TRL using these components were also important objectives. During the ≈ 18 hr PWM-5 operations, and depending on how they are counted, ≈ 839 crew setpoint interactions were recorded achieving ≈ 51 E&F tests, ≈ 31 Hydroponics tests (without bubbles), ≈ 18 Hydroponics tests with Aeration, Bubble Diversion, Bubble Separation, and Water Trap, ≈ 76 Bubble Diversion tests, and ≈ 38 Bubbles Separation tests. ‘Deeply stable’ steady, steady periodic, stable aperiodic, and unstable modes of operation were demonstrated. Hundreds of thousands of oxygenating bubbles were generated and separated. Flow conditions were established under which 100% phase separations were achieved for BS, WT, BD, and TC for various root models. The limits of operation of these capillary fluidic devices are in part established by the PWM-5 tests. The high repeatability of E&F function despite the variability on root zone density and geometry is demonstrated. Dual passive reservoir infill was also demonstrated.

PWM-6 Performance Summary

The PWM-6 demonstrations seek to quantify the impact of TC pinning, contact angle hysteresis, and TC flow resistance due to the presence of the root models on the ability to achieve stable parallel hydroponic flow. The stable long duration performance of capillary devices, BD, BS, and WT are also sought. Increased confidence and TRL are sought with significantly increased runtime for the system. The $\approx 18:45$ hr PWM-6 operations completed ≈ 90 E&F tests, 5 bubbly E&F tests, ≈ 50 steady Hydroponics tests with as much as ≈ 30 min maintained without change. Parallel channel flow instabilities are readily observed. Additional limits on device performance were demonstrated for parallel TC flow, serial TC flow, TC gas ingestion, E&F, BS, WT, and BD as functions of root model, TC fill level, and pump flowrate. Specific tests were performed at increased camera zoom, for TC, BS, BD, tubing fittings, and TC pinning edge, lid and stem regions.

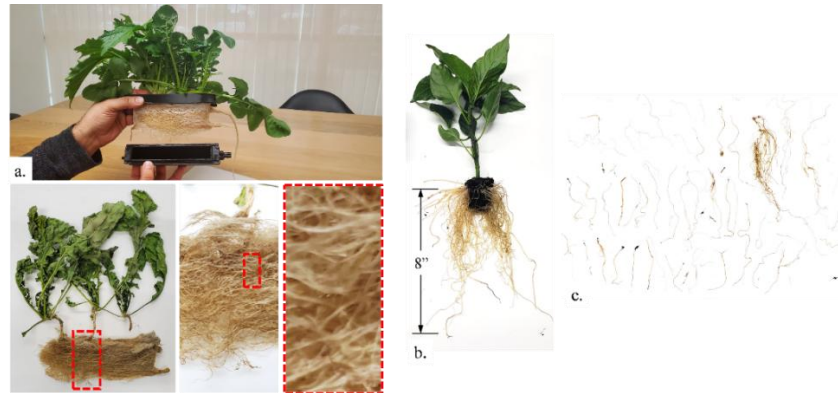


Figure 4. Early hydroponic ground growth tests using the PWM Hydroponic TC to aid selection of synthetic root models: a. Approximately 40-day arugula extracted from TC with magnified root zone images. b. 45-day pepper root zone removed from TC and c. dissected to record root diameters, lengths, and length distribution.

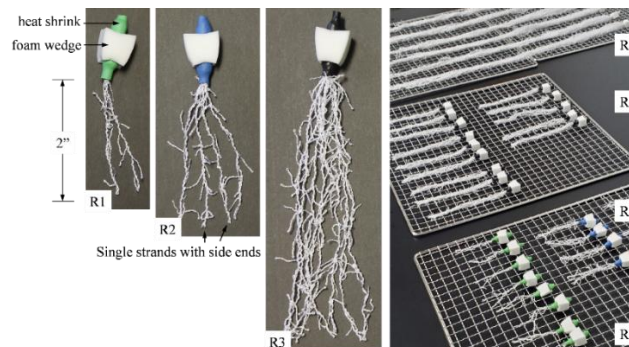


Figure 5. Labeled images of synthetic flight root models R1, R2, R3, and R4, intending to model sparse and packed root limits (Ref. Table 1).

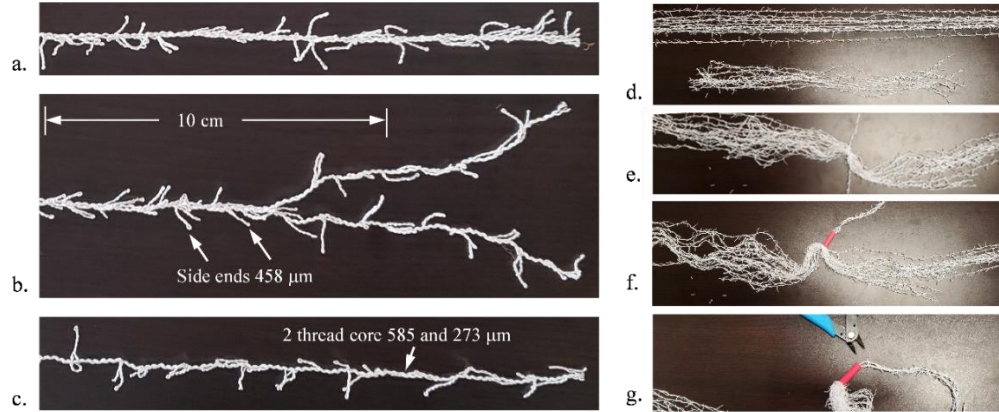


Figure 6. Synthetic roots and assembly. a. As purchased: double-wound strand, b. split, and c. as employed as single strand (ODs labelled). Root model R4 assembly: d. layout, e. center, overlay and tie off, f. draw through polyolefin heat shrink, and g. cut to finish.

PWM-5b Performance Summary

The PWM-5b operations exploited the broadly written PWM-5 Crew Procedures to further investigate flow regime limits of the PWM hardware at higher camera resolution, establish ingestion limits within the TC for all root types R0-R4, and demonstrate the robust response of the system to a broader, less controlled, range of gas-liquid configurations. High resolution performance of the WT was also pursued. The $\approx 18:50$ hr PWM-5b operations completed include ≈ 48 E&F tests and ≈ 93 shorter duration (~ 5 min) steady single liquid phase Hydroponics tests. Limits on device performance were demonstrated for the capillary fluidic devices of the system with ≈ 68 tests performed for the combined TC, BS, WT, and BD system, ≈ 76 tests specifically for the BD, and ≈ 38 for the BS. A cavalier pump-driven system priming was conducted quickly without regard for trapped bubbles since bubbles are passively 100% separated by plumbing elements of the system. Specific WT tests were completed in haste, but successfully demonstrated water trap, bubble separation, steady two-phase flow, and static drain function. Without consultation from the ground team, crew member Michael Barratt applied the ‘over-the-shoulder’ camera as an orthogonal zoomed in camera effectively doubling our science footage. This capability led to our immediate change in plan to use one camera for zoomed high-resolution and the other for our original FOV. Again, without consultation, crew member Matthew Dominick configured lighting to resolve fine bubble motion to accurately quantify bubble separation regimes for TC, BD, BS, and WT.

V. Combined Assessment of PWM 5 & 6 Objectives

In total, for PWM-5, -6, and -5b, ≈ 189 E&F tests, ≈ 124 steady Hydroponic flows, and ≈ 306 phase separating limits tests of TC, BS, WT, and BD were conducted. Bracketed by the established limits of operation, the success of the hardware to perform the desired passive functions establishes approximately TRL 7 for the system as well as the various components: TC, Aerator, Reservoirs, Pump, BS, WT, and BD. From a user perspective, mundane plant installation, prime, hydroponic flow $1 \leq Q \leq 5$ mL/s with and without aeration, passive reservoir infill for water and nutrient solution, and easy plant placement and removal are successfully demonstrated by PWM -5, -5b, and -6. In the remainder of this document we report on the PWM data archive, selecting a variety of test results from it that can be applied to advanced system design. Cross-cutting applications of the capillary fluidic demonstrations are numerous.

VI. Data Archive in Work

In total, the PWM-Hydroponics 5, 6, and 5b experiments produced ≈ 55.5 hr of video footage (≈ 100 hr including over-the-shoulder camera). The video data was received from NASA in hundreds of clips which were then carefully spliced into single continuous .mp4 files for each day of operations. Depending on the day, HD or 4K videos were recorded. In the case of GMT285 PWM-5b, the day’s video is split into two spliced videos for Standard HD and 4K portions. The videos were time-stamped to approximate GMT ± 1 s.

During the operations, chronological experimental operations are compiled at 1 Hz in Log Files tracking GMT, Crew, Test Operation, Pump Setting, TC (up to 2 ea), Plant Root Model (up to 4 ea), Syringe state (Vol., up to 4 ea), PV state (up to 9 ea), CV state (up to 4 ea), SC state (up to 3 ea), and Res state (up to 2 ea). Many hundreds of

thousands of data points were tabulated for each PWM day of operations in this way. The large Log Files are condensed into Summary Tables that include only events for which a significant change to the state of the system was made. The Summary Tables are used to sort PWM operations such as Test Names Prime (PR), E&F tests (EF) and Hydroponics Flow (HF), Hydroponics flows with Aeration (AE), BS, WT, BD, etc. The various events are hyperlinked to the source video corresponding to the GMT of the event. The Data Log files and Operations Summary spreadsheets are included with all GMT timestamped video records in the PWM 5 & 6 archive (<https://psi.nasa.gov>).

VII. Selection of Data Reduction

The large PWM 5 & 6 dataset is enabled by the breadth of discrete and continuous variables achievable by the PWM hardware. For example, initial conditions can vary widely depending on the nature of the initially primed states of the Fruit Punch Drink Bag, 2 ea 120 mL Syringes, PWM Conduit, Reservoirs, and TC (R0-R4). The E&F tests are functions of root model R0-R4, Syringe and Pump E&F rate, and gas ingestion/injection E&F characteristics. The serial and parallel Hydroponic tests are performed for a continuous variety of liquid fill levels and pump flowrates for the various modes of operation: single-phase liquid flow and two-phase gas-liquid flow with Aeration and BD, Aeration and BS, WT, and BD—with and without air injection (30 mL Syringe). Isolated limit tests are also performed on the Aerator, BS, WT, BD and TC. Data reduction items for root models R0-R4 include System Primes, E&F, Serial and Parallel Hydroponic Flows, Devices Performance Limits and Regimes Maps, and Diagnostics. Only E&F, and Serial Hydroponic flow tests are briefly addressed herein.

A. E&F Tests

Most of the E&F states are provided in Figures 7-109 for PWM-5, -5b, and -6, respectively. These images provide clear visual comparisons of the repeatability of the E&F process, which depends mostly on the root model geometry for essentially fixed TC and flowrate 10 to 20 mL/min. We note that there is no significant disturbance to the root models at these E&F flowrates. E&F flowrates and average flow times are readily determined from the video footage with corrections accounting for gas injected/ingested into/out of the TC, conduit, and 120 mL Syringe. Uncertainties in the volume measurement are approximately $< \pm 2$ mL. A more detailed comparison of PWM-5 E&F data from Figure 7 is provided for select back-to-back tests in Figure 8. Typical hold-up volumes are listed in Table 3 for PWM-5. The repeatability of the E&F states is clear. Initial and final Ebb states are shown in Figure 9 for PWM-5 with R4, between which ≈ 6 hr of Hydroponic Flow testing and overnight dormancy (≈ 12 hr) has transpired. The states remain highly repeatable with similar E&F fill and hold-up volumes as well as visual appearance. From Figure 7 we observe that E&F volumes can be significantly reduced when conducted through TC inlet or outlet ports, the effect of which amplifies with increasing root volume. Poorer Ebb performance is expected from the side ports due to the plant geometry and longer aspect ratio of the draining process. In certain Ebb cases, gas ingestion was nearly immediate during Ebb leaving nearly the entire liquid contents in the Root bundle within the TC (i.e., R4 inlet, Fig. 7). In general, however, despite poorer performance, E&F behavior through TC lower inlet and outlet remains highly repeatable.

E&F tests were conducted where air is ingested either purposefully or inadvertently through the TC base during the Ebb cycle but purposefully rejected during the Flow cycle by re-routing the Flow through the BS, BD, and TC inlets such that nearly 100% liquid Flows are achieved no matter what occurs during Ebb. In this manner is further control offered by demonstrating high liquid content Flow despite highly two-phase Ebb. As examples, slugs and bubbles ingested during Ebb and eliminated during Flow are repeatably demonstrated for PWM-5b with R4 (GMT284-15-24:01:17:00). Large gas ingestions reaching the 120 mL Syringes require gas volume balancing to determine accurate liquid E&F volumes, but most of the gas ingested during Ebb is removed by the BS, BD, and TC during Flow. Many other examples may be observed (i.e., PM-6, R4, TC 1, GMT198-21-13-1298:24:14 and PWM-6 GMT199-22-03-1307-part2:52:50). A related extreme event is captured during an inadvertent air ingestion during PWM-6 tests when the 120 mL Syringe 3 plunger was retracted while PV-2, PV-4, PV-5, and PV-8 were open. Large gas slugs were ingested through the BS liquid and gas outlet lines and into the syringe (GMT198-21-13-1298:26:02)—a model for worst case Ebb ingestion. Excellent recovery via elimination of the ingested gas is demonstrated during the subsequent Flow through BS, BD, and TC.

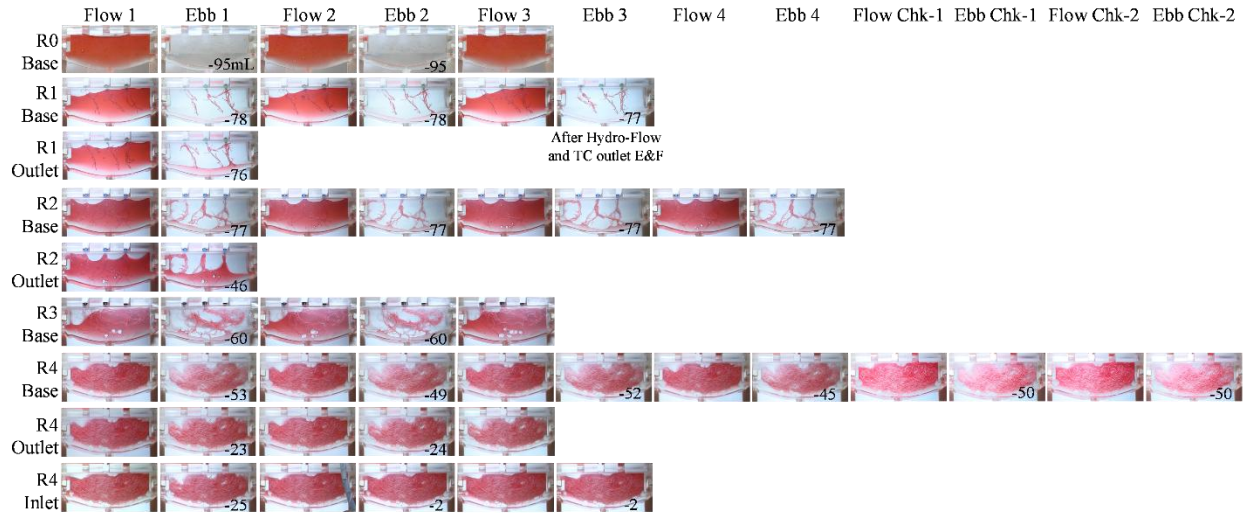


Figure 7. Example of repeat before and after E&F tests for PWM-5 for different Root Models and from different TC ports. Ebb volumes are listed in mL. Spot Check E&F tests are performed after lengthy Hydroponic flow tests and then overnight showing repeatability of the E&F process for Root Models, R0-R4.

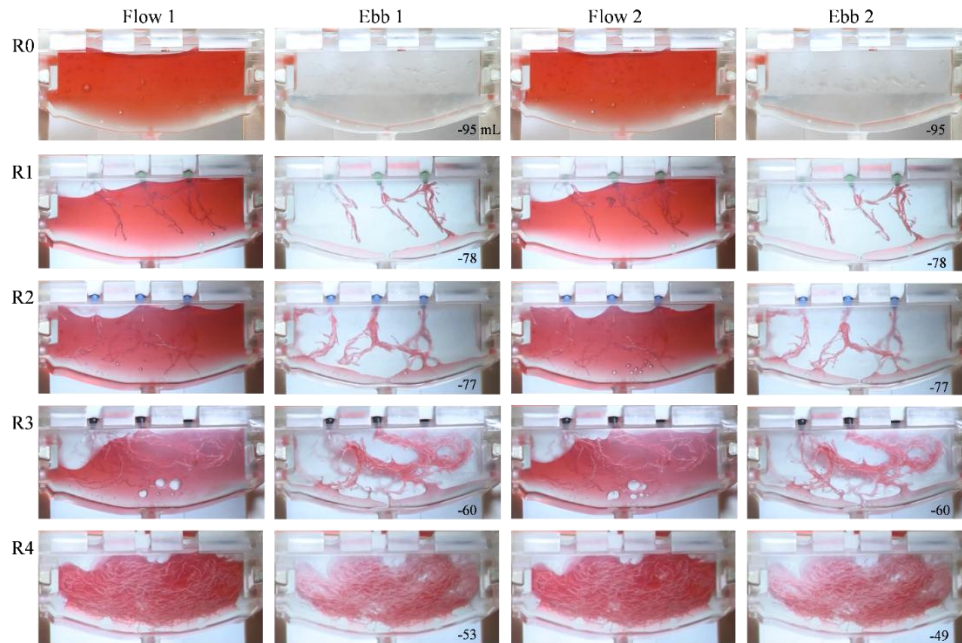


Figure 8. Select back-to-back E&F states for PWM-5 from Figure 7 for R0-R4 draining from TC base and identifying liquid volume removed during Ebb prior to ingestion. (Ref. to Table 3 for Hold-up values.)

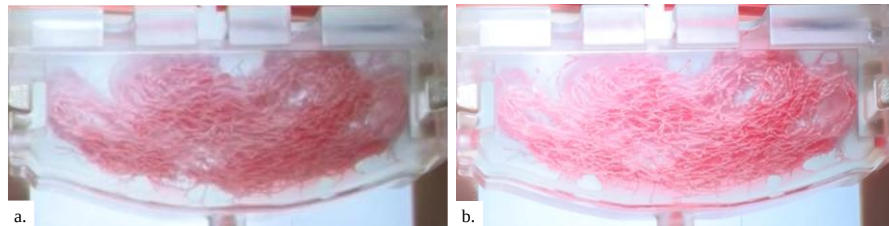


Figure 9. a. First and b. last Ebb state for PWM-5 following ≈ 6 hr of Hydroponic Flow testing and overnight dormancy (≈ 12 hr). The states remain highly repeatable.

Table 3. PWM-5 E&F root model hold-up and root model volume displacements. Note that the hold-up values are for liquid only and that root model displaced volumes for R0, R1, and R2 are negligible.

Root Model	Hold-up (mL)	Root Volume (mL)
R0	< 1	0.0
R1	17	0.1
R2	18	0.3
R3	34	1.2
R4	39	5.2

Summarizing the ≈ 200 E&F demonstrations for PWM-5, -5b, and -6 we note that for a given root geometry and TC, the E&F process is highly repeatable, particularly when conducted through the most effective TC base port. We find that Ebb volume decreases by up to $\approx 50\%$ between R0 and R4, with $> 90\%$ contributions due to liquid hold-up within the root bundle and $< 10\%$ due to root bundle displacement (root volume). Some increase in Ebb volume is expected with significantly lower drain rate, but drain port geometry, local drain port region root geometry, and TC aspect ratio play significant limiting roles of the degree to which the various root models can be drained.

B. Hydroponic Flow: Aeration, BD

Any plant watering system for application in space must account for the presence of bubbles whether produced *inadvertently* by trapped gas, degassing, or biochemistry, or *purposely* by bubble ingestion or direct injection for liquid aeration. Because inadvertent bubbles have the potential to completely destabilize such capillary flows, and because plants in low-g environments often suffer from hypoxic conditions, PWM 5 & 6 dedicated over 7 hours of experiment time to investigating bubbly flows in the loops. At least 129 PWM-5, -5b, and -6 tests were conducted towards this end. In general, we find that the PWM hardware is largely insensitive to the presence of bubbles due to the redundant passive bubble separating devices (BS, BD, and TC/Root Model). A host of destinations for such bubbles are observed. With all devices plumbed, $\approx 100\%$ bubbles are separated by the BS. Those that ‘pass’ are diverted by the BD into the upper entrance region of the TC. These bubbles may become temporarily trapped in recirculation zones within the TC created in part by the presence of the plant roots. They may be trapped by such roots, become wall-bound, grow by mergers until confined, and then rise in the channel by capillary forces only to coalesce with the free surface and leave the flow. Bubbles < 1 mm circulate through the loop, but even these appear to eventually become trapped, wall-bound, dissolve, or coalesce into larger bubbles which are then separated by the next device downstream. We note that no bubbles are observed to escape the TC when packed roots are present (i.e., R4).

The BD was designed with the intention to improve the gas separation efficiency of the TC by introducing the bubbles close to the TC free surface. The BD exploits its own bubble point allowing only liquid to pass into the lower TC inlet, while excluding bubbles and directing them into the upper TC inlet. To demonstrate the performance of the BD, the Aerator is employed producing variable trains of bubbles upstream of the pump, or the AIS is employed injecting variable bubble trains downstream of the pump. TC fill level, pump flowrate, flowrate ratio, and Root Models are varied during such tests representing all independent parameters for the BD demonstrations and such that BD separation efficiency may be quantitatively established in a regime map. We provide only a terse display of such data herein.

Example still images are presented in Figure 10 for Hydroponic flow with Aeration and BD operation. The figure selects cases from PWM-5b with R4 in the TC for increasing Aeration rates and fixed pump flowrate. In Figures 10b-d, zoomed 4K video reveals 100% sub-liminal diversion, 100% liminal diversion, and $> 95\%$ super-liminal diversion, respectively. These conditions are created by increasing the degree of aeration for fixed pump flowrate. As best we can detect, 100% BD diversions assure 100% liquid delivered to the TC base assuring and thus $\approx 100\%$ gas separation in and by the TC/Root Model system. The diverter limit in this instance is observed when bubble densities increase to the point of a coarsening ‘foam’ of increased viscous resistance. The increasing ΔP eventually exceeds the bubble point of the BD screen leading to aperiodic gas breakthrough at the critical point. The gas content at breakthrough increases with increasing gas content and pump flowrate. Below such flow limits identified during PWM testing, the BD combined with the TC and Root Model serve the role of 100% separation of a wide variety of rogue or purposely aerating bubbles. The PWM BD is currently assessed at TRL 7.

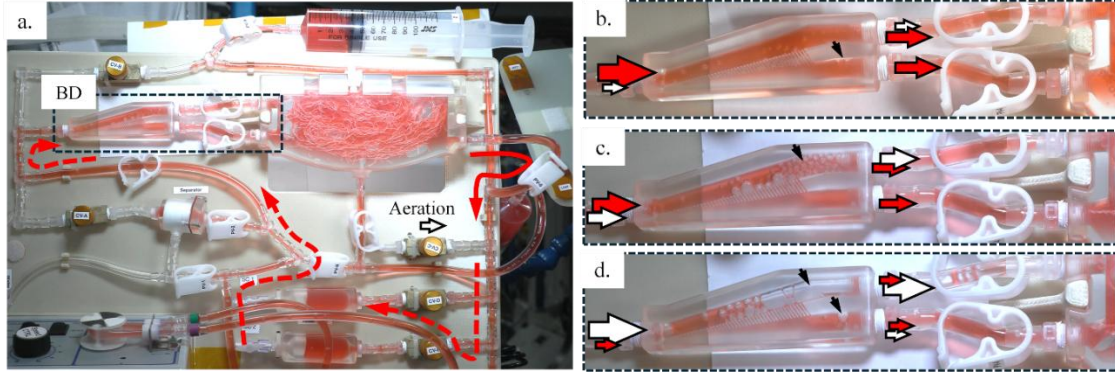


Figure 10. a. PWM-5b Hydroponic flow with Aeration and Bubble Diversion (black dashed region) with solid red arrows indicating single phase liquid flow, white arrow Aeration flow, and dashed red arrows 2-phase flow. Close up images of BD during tests of increasing gas flow ratio: **b.** 100% gas diversion mode, **c.** liminal 100% gas diversion mode, and **d.** super-liminal mode where gas flowrate exceeds BD bubble point leading to gas breakthrough with flowrates notionally indicated by arrow size. Small black arrow in **b.** identifies stationary wall-bound bubbles, **c.** bubble/foam confinement and back-up, **d.** bubble coarsening and breakthrough.

C. Hydroponic Flow: Aeration, BS, WT, BD

In a similar manner to the Hydroponics tests with Aeration and BD, numerous Hydroponic flow demonstrations of passive bubble separation were conducted for Aeration, BS (with WT), BD and TC with Root Model. Flowrate variation along with Aeration or gas injection through the AIS are used to map BS conditions of 100% separation. A brief presentation is provided in Figure 11 where the general plumbing circuit is shown with select magnified images of BS performance illustrating the approach to the bubble point limits of operation from the 100% phase separation regime. These limits are found in part by turning down CV-A increasing the pressure within the BS to the point of liquid breakthrough. Tests performed beyond the breakthrough limit are performed to demonstrate 100% liquid trap and separation in the WT to be discussed shortly. Two-phase flow regimes (i.e., flowrates and flowrate ratios) are also varied to identify bubble carry-over limits of the BS device, where ≤ 1 mm diameter bubbles are first observed to escape the BS liquid outlet line. Hundreds of BS tests were performed, which may be used to construct BS limiting performance regime maps. Such limits were observed and will be included in subsequent maps as the data in the archive is further reduced. We observed a small but certain degradation in BS performance after hours of operation which will receive special attention during further data reduction efforts.

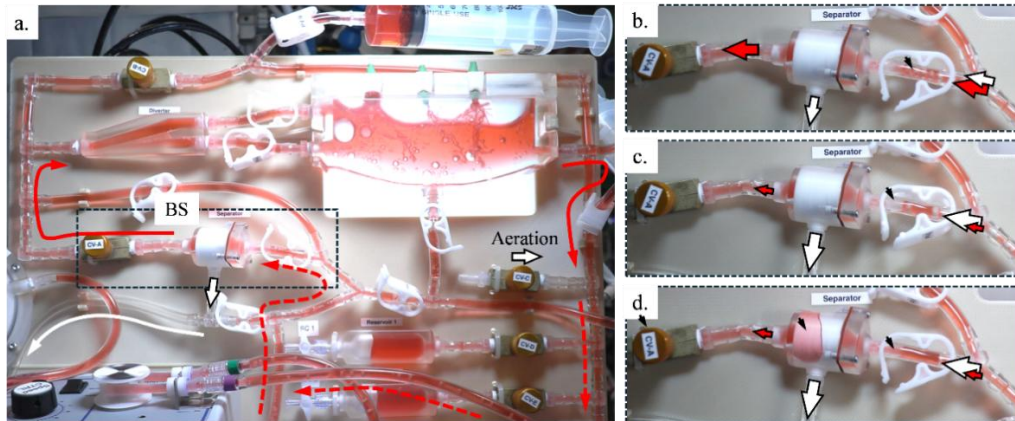


Figure 11. a. PWM-5 Hydroponic flow tests with Aeration, BS, BD, and TC with R1—100% gas separation is achieved by the BS with 100% gas (white arrow) vented to atmosphere and 100% liquid (solid red arrow) returning to loop. Dashed red arrows denote two-phase flow. Zoomed 4K video of BS performance varying flowrate, gas content, and system pressure: **b.** 100% separation for low gas content bubble flow (black arrow), **c.** 100% separation for high gas content gas slug flow (black arrow), and **d.** bubble point liquid breakthrough limit (black arrow) for a gas slug flow (black arrow) achieved by increasing downstream pressure via turning down CV-A (black arrow).

VIII. Discussion and Summary: Overall PWM Performance from an Applications Perspective

From the PWM 5 & 6 demonstrations, the E&F technique is highly repeatable for plants in the short term (< 1 day), but strongly dependent on changes to root zone geometry in the long term as the plants mature (> 1 day). Reductions to E&F volumes of > 50% over the course of plant maturation can be expected due mostly to increasing liquid hold-up within the root zone and to a much lesser extent increasing root volume. The liquid volume lost to evapotranspiration during each E&F cycle is the difference between Flow and Ebb volumes in the short term. Unless long E&F cycles are employed, E&F volumes reduce over time in the long term despite increasing losses to evapotranspiration as the plant matures. Furthermore, large hold-up volumes may lead to hypoxic conditions if the nutrient solution is not oxygenated. The microgravity E&F phenomena is essentially bubble-free. E&F tests where air was purposely or inadvertently ingested during the Ebb cycle through the TC base was passively separated (via BS, BD, TC) for 100% liquid fill through the TC inlet during the Flow cycle. Such capabilities provide further E&F control options, limiting liquid/gas volume ratcheting and concerns of root zone dry-out or TC over-fill. The capillary fluidics design guides employed for the PWM 5 & 6 TC development are adequate to assure stable static and dynamic states for a successful E&F technique aboard spacecraft—despite such a poorly wetting aqueous solution.

The passive capillary fluidic phenomena reported in this paper, as directed to plant watering aboard spacecraft, contribute to future space missions by expanding levels of passive fluidic control akin to terrestrial hydroponic and E&F systems by replacing the role of gravity with the combined roles of surface tension, wetting, and system geometry, which are in turn exploited for more robust no-moving-parts systems less sensitive to potentially disastrous bubbles, and illuminating a clearer path to full automation. Because the present PWM approach behaves similarly as if employed on earth, it may be reasonable to expect similar performance for practical hydroponic cultivation and contamination (i.e., from plant roots, algae, fungi, etc.). The employment of real plants one day may tell a different story.

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